Quantifying Fortune for Table Tennis Tournaments

Applied to the Paris 2024 Olympic Men's Singles Event

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Abstract

This paper develops a quantitative framework for measuring tournament draw favorability in professional table tennis. Using 390 clean matches from nine pre-Olympic tournaments, we estimate player strength via a three-feature model combining recent win rate, ITTF ranking score, and career achievement tier. Applied to the Paris 2024 Olympic bracket, the model produces a Draw Fortune Index measuring whether each player's bracket path was easier or harder than their seed predicts. Seeding explains only 17.3% of draw difficulty ($R^2 = 0.173$). Fortune correlates significantly with final placement (Pearson $r = -0.753$, $p < 0.001$; Spearman $r = -0.601$, $p < 0.001$). The luckiest player was Dang Qiu (Fortune = $+0.752$); the unluckiest was Truls Möregardh (Fortune = -2.081), who reached the final despite the hardest draw in the field.

Contents

1 Introduction

In single-elimination tournaments, outcomes depend on player ability and on the sequence of opponents determined by the draw. A top seed may face elite opponents early while a lower seed advances through a weaker bracket.

The Paris 2024 Olympic table tennis men's singles is a clear example. Wang Chuqin, the world's top-ranked player (Score = 0.917), was eliminated in the second round. Fan Zhendong (Score = 0.737) won gold. This paper asks: did the draw, not skill, determine these outcomes?

Three questions guide the work:

1. Can a three-feature model reliably estimate player strength before the Olympics?
2. Can draw path difficulty be quantified into a Draw Fortune Index?
3. How accurately do predicted outcomes match actual results, round by round?

This work connects to tournament theory [2], the Bradley-Terry model [1], and mechanism design [3].

2 Theoretical Background

2.1 Tournament Theory and Mechanism Design

Tournament theory [2] shows that competition structure affects outcomes independently of ability. Mechanism design asks whether structures produce fair outcomes. The Draw Fortune Index tests whether Olympic seeding equalizes draw difficulty as intended.

2.2 The Bradley-Terry Model

Win probability for player i against j:

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j}$$

Our three-feature score estimates π_i. All predicted match probabilities are computed from this formula.

2.3 Zero-Sum Games

Every match is zero-sum. The bracket is a sequence of such games where one loss ends a player's tournament, amplifying the effect of any upset.

3 Data

3.1 Tournament Match Data

3.2 Supplementary Data

Two additional datasets support the strength model: ITTF World Rankings (Week 30, July 23 2024) and a manually constructed career achievement tier table (scores 1 to 7). The Paris 2024 Olympic bracket was sourced from Wikipedia.

Table 1: Tournaments in dataset

Tournament	Matches
Singapore Smash 2024	63
Saudi Smash 2024	63
World Cup Macau 2024	15
WTT Champions Incheon 2024	31
WTT Champions Chongqing 2024	30
WTT Star Contender Doha 2024	47
WTT Star Contender Goa 2024	47
WTT Star Contender Ljubljana 2024	47
WTT Star Contender Bangkok 2024	47
Total (after cleaning)	390

3.3 Data Cleaning

All names standardized to uppercase. BYE rows removed (390 matches remain). WinnerName derived from the Winner (1 or 2) column. A separate per-player win-rate table was built from these 390 matches; players with fewer than three matches were excluded from that table only, so opponents' records in the underlying match data remain intact. Six name mismatches across datasets (e.g., *Jang Woo-Jin* vs. *Jang Woojin*, *Wang Yang* vs. *Yang Wang*) were resolved manually. Players absent from tournament data received median feature values. Final dataset: 390 matches, 74 players with win rate data, 49 Olympic players modeled.

One additional data-quality note: 16 of the 19 players eliminated in the Round of 32 had no recorded opponent in the source bracket file (only their automatic Round-of-64 bye was listed). For these players, Draw Difficulty was imputed using the median player Score rather than a specific opponent's score; see the limitations discussion in Section 8.

4 Three-Feature Strength Model

4.1 Feature 1: Recent Win Rate

$$\text{WinRate}_i = \frac{\text{Wins}_i}{\text{Matches}_i}$$

Captures current form from the nine pre-Olympic tournaments.

4.2 Feature 2: ITTF Ranking Score

$$\text{RankScore}_i = \frac{\text{Points}_i}{\max_j(\text{Points}_j)}$$

Wang Chuqin held 7925 points (RankScore = 1.000). Captures long-run accumulated performance.

4.3 Feature 3: Career Achievement Score

Normalized: $\text{AchievScore}_i = \text{Tier}_i/7$.

Table 2: Career achievement tiers

Tier	Achievement	Example
7	Olympic Champion	Fan Zhendong
6	World Champion / Olympic Silver	Wang Chuqin, Möregardh
5	Olympic Bronze	Félix Lebrun, Ovtcharov
4	Olympic 4th / Team Gold	Calderano, Jang Woo-Jin
3	World SF / European Champion	Harimoto, Dang Qiu
2	World QF / European medal	Mid-tier players
1	National level only	Lower seeds

4.4 Combined Score

$$\text{Score}_i = \frac{1}{3}\left(\text{WinRate}_i + \text{RankScore}_i + \text{AchievScore}_i\right)$$

Table 3: Top 15 players by strength score

Rank	Player	WinRate	RankScore	AchievScore	Score
1	Wang Chuqin	0.895	1.000	0.857	0.917
2	Fan Zhendong	0.750	0.462	1.000	0.737
3	Félix Lebrun	0.688	0.453	0.714	0.618
4	Hugo Calderano	0.643	0.447	0.571	0.554
5	Dimitrij Ovtcharov	0.556	0.194	0.714	0.488
6	Tomokazu Harimoto	0.667	0.332	0.429	0.476
7	Dang Qiu	0.692	0.254	0.429	0.458
8	Jang Woo-Jin	0.583	0.210	0.571	0.455
9	Lin Yun-Ju	0.583	0.336	0.429	0.449
10	Cho Dae-Seong	0.625	0.150	0.571	0.449
11	Truls Möregardh	0.333	0.127	0.857	0.439
12	Darko Jorgic	0.444	0.153	0.429	0.342
13	Sharath Kamal Achanta	0.600	0.081	0.286	0.322
14	Ovidiu Ionescu	0.600	0.073	0.286	0.319
15	Omar Assar	0.375	0.135	0.429	0.313

Truls Möregardh (Rank 11) illustrates the model's value: his low WinRate (0.333) and RankScore (0.127) are offset by a high AchievScore (0.857), giving a more balanced estimate than win rate alone.

5 Predicted Match Outcome Matrix

$$\hat{P}(i \text{ beats } j) = \frac{\text{Score}_i}{\text{Score}_i + \text{Score}_j}$$

Each cell is W ($\hat{P} > 0.5$), L ($\hat{P} < 0.5$), or D ($\hat{P} = 0.5$), producing a 49×49 predicted outcome matrix before the Games. Head-to-head data was tested as an override but excluded: only 5 of 37 actual Olympic matches had H2H records, and Bradley-Terry matched or outperformed H2H on those 5 cases.

Note: this section's predicted-accuracy figures (round-by-round accuracy and the upset analysis in Section 7.2–7.3) reflect the original analysis and have not yet been re-verified against the rebuilt 390-match dataset described in Section 3. They should be re-run before final submission.

6 Draw Difficulty and Fortune Index

6.1 Draw Difficulty

$$\text{Difficulty}_i = \sum_{r \in \text{Rounds}} \text{Score}_{\text{opp}(i,r)}$$

For each round a player competed in, the Score (Section 4) of their actual opponent is added. BYEs are excluded since no real opponent is faced. A higher total means the player walked a harder bracket path.

6.2 Expected Difficulty

Difficulty alone cannot say whether a draw was unfair, since top seeds are designed to face stronger opponents than low seeds. To isolate the unfair component, we first estimate how much Difficulty a player's seed alone would predict, using ordinary least squares (OLS) linear regression:

$$\text{ExpectedDifficulty}_i = \hat{\alpha} + \hat{\beta} \cdot \text{Seed}_i$$

where the coefficients come from the observed pairs $(\text{Seed}_i, \text{Difficulty}_i)$ across all 49 modeled Olympic players. OLS fits the line minimizing the sum of squared vertical distances:

$$(\hat{\alpha}, \hat{\beta}) = \arg\min_{\alpha,\beta} \sum_{i=1}^{49} (\text{Difficulty}_i - \alpha - \beta \cdot \text{Seed}_i)^2$$

This produces $\hat{\alpha} = 0.981$ and $\hat{\beta} = -0.0158$ ($R^2 = 0.173$). The negative slope confirms that, on average, low seed numbers (stronger players) faced higher Difficulty and high seed numbers faced lower Difficulty, consistent with how Olympic seeding is designed to work.

6.3 Draw Fortune Index

$$\text{Fortune}_i = \text{ExpectedDifficulty}_i - \text{Difficulty}_i$$

Positive Fortune means the player's actual path was easier than their seed predicted (lucky); negative Fortune means it was harder (unlucky).

7 Results

7.1 Draw Fortune Index

Fortune correlates significantly with final result (Pearson $r = -0.753$, $p < 0.001$; Spearman $r = -0.601$, $p < 0.001$). Players with harder draws tended to advance further, reflecting the seeding system's design: stronger players are placed into harder bracket halves.

Five of the ten most-fortunate players in Table 4 (Marcos Freitas, Quadri Aruna, Cho Dae-Seong, Edward Ly, and others not shown) reflect the median-imputation step described in Section 3.3 rather than a specifically known easy opponent, since their actual Round of 32 opponent was not recorded in the source bracket file. Their Fortune values should be read as approximate. Dang Qiu's result is not affected by this imputation and is the most reliable data point in the table.

Table 4: Most and least fortunate players, Paris 2024

Seed	Player	Difficulty	Exp. Diff.	Fortune	Result
Most fortunate					
7	Dang Qiu	0.118	0.871	+0.752	2nd round
12	Marcos Freitas	0.221	0.791	+0.571	1st round
14	Quadri Aruna	0.221	0.760	+0.539	1st round
15	Cho Dae-Seong	0.221	0.744	+0.523	1st round
22	Edward Ly	0.221	0.633	+0.412	1st round
Least fortunate					
19	Truls Möregardh	2.762	0.681	−2.081	Silver
3	Félix Lebrun	2.462	0.934	−1.528	Bronze
59	Kanak Jha	1.179	0.048	−1.131	Round of 32
2	Fan Zhendong	1.705	0.950	−0.755	Champion
39	Yang Wang	0.917	0.364	−0.553	1st round

7.2 Round-by-Round Accuracy

[This subsection, including Tables 5–8 and the upset analysis from the original draft, is pending re-verification against the rebuilt dataset and is omitted here to avoid presenting unverified figures as confirmed results. See the note at the end of Section 5.]

7.3 Key Match Analysis

Wang Chuqin vs. Truls Möregardh (Round of 32). The tournament's defining upset. Wang (Score = 0.917) was the heavy favorite by the three-feature model but lost. Möregardh (Fortune = −2.081) then advanced through the hardest recorded draw in the field to reach the final, losing only to eventual champion Fan Zhendong.

Dang Qiu (Score = 0.458, Fortune = +0.752) lost in the second round despite the easiest draw, by Fortune, of any player in the field. His result illustrates the paper's key finding: favorable draws do not guarantee advancement.

8 Discussion

Main findings:

1. Seeding explains only 17.3% of draw difficulty ($R^2 = 0.173$). Over 80% of variation is structural randomness, showing the draw mechanism introduces meaningful inequality beyond seeding rank.

2. Fortune is a significant predictor of results (Pearson $r = -0.753$, $p < 0.001$), confirming that bracket path plays a measurable role independent of player quality.

3. Möregardh's run (Fortune = −2.081) is the tournament's defining anomaly: the hardest structural path combined with a run to the final.

4. Dang Qiu's exit (Fortune = +0.752) shows that the easiest recorded draw in the field did not translate into a deep run, reinforcing that Fortune measures opportunity, not outcome.

Limitations: The Round of 32 opponent was unrecorded for 16 of 19 players eliminated in that round; their Difficulty was imputed using the median player Score rather than a specific opponent, so their individual Fortune values carry more uncertainty than players with fully recorded paths. Nine of the 49 Olympic players had no win-rate data in the nine pre-Olympic tournaments and received the median WinRate. World Cup Macau, WTT Champions Incheon, and WTT Champions Chongqing yielded fewer matches in the available source files than the original tournament totals would suggest, slightly limiting win-rate precision for players active mainly in those events. Equal feature weights were not optimized. The round-by-round prediction accuracy analysis (Section 7.2) requires re-verification against the rebuilt dataset before its figures can be reported as confirmed.

Future work: Re-verify the predicted match outcome matrix and round-by-round accuracy against the rebuilt 390-match dataset. Optimize feature weights via logistic regression. Build a probabilistic opponent tree. Extend to women's singles and prior Olympic cycles.

9 Conclusion

A three-feature player strength model and a Draw Fortune Index were constructed for the Paris 2024 Olympic table tennis men's singles using a fully reproducible data pipeline. Seeding explains only 17.3% of draw difficulty, and Fortune correlates significantly with final placement (Pearson $r = -0.753$, $p < 0.001$). Truls Möregardh, the most structurally disadvantaged player by this measure, produced the tournament's most remarkable performance by advancing to the final from the hardest recorded bracket path in the field, while Dang Qiu's early exit despite the easiest recorded path underscores that a favorable draw is an opportunity, not a guarantee.

References

[1] Bradley, R.A. and Terry, M.E. (1952). Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons. *Biometrika*, 39(3/4), 324–345.

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[4] Elo, A.E. (1978). *The Rating of Chessplayers, Past and Present.* Arco Publishing, New York.